TECHNICAL ASSESSMENT REPORT

SECURIAGENT AI • LOG ANALYZER & COMPLIANCE ENGINE

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Company: Kapstone Technological Services

EXECUTIVE SUMMARY

This report details the implementation of **SecuriAgent AI**, an end-to-end security log analysis tool designed to help Security Operations Center (SOC) teams identify and respond to cyber threats. The solution automatically reads server logs, detects strange activity using machine learning, groups related security alerts, maps issues to standard compliance frameworks, and provides actionable reports.

1,000	111	52
LOGS ANALYZED	ANOMALIES DETECTED	HIGH-RISK ASSETS

TASK 1: DATASET PREPARATION AND MODEL TRAINING

To build and test the system, I created a clean, realistic cybersecurity log dataset containing 1,000 log records. The dataset includes normal system traffic (like typical web browsing and successful logins) as well as malicious activity (such as brute-force password guessing, port scanning, and SQL injection attempts).

- **Data Preprocessing:** Key numerical features were extracted: login failure count, bytes transferred, and status codes. These features were scaled using `StandardScaler`.
- **Model Selection:** I selected an **Isolation Forest** model, which is an unsupervised machine learning algorithm ideal for finding rare anomalies in network traffic.
- **Performance & Limitations:** The model successfully identified suspicious spikes in activity. Main limitations include occasional false positives from routine admin tasks, which are addressed during alert correlation.

> **Key Insight — Task 1**
> Using an unsupervised anomaly detection model allows SecuriAgent AI to spot brand-new, unseen threat patterns without relying strictly on fixed, pre-written rules.

TASK 2: SECURITY LOG ANALYSIS AND ALERT CORRELATION

Single suspicious logs can cause false alarms. To prevent this, SecuriAgent AI uses an alert correlation engine that groups individual alerts based on source IP address, event type, and timestamp windows. This combines multiple related warnings into a single, structured security incident.

TASK 3: COMPLIANCE MAPPING

Detected security threats are automatically mapped to standard governance frameworks (**ISO 27001** and **NIST CSF**) to ensure immediate compliance readiness and easy remediation:

Threat Event	NIST CSF Control	ISO 27001 Reference	Risk Level	Recommended Remediation
LOGIN_FAILURE	PR.AC-1 (Access Control)	A.9.4.2 (Secure Log-on)	HIGH	Enforce lockout rules after 5 failed attempts.
PORT_SCAN	DE.CM-1 (Network Monitor)	A.13.1.1 (Network Controls)	MEDIUM	Block source IP on perimeter firewalls.
SQL_INJECTION	PR.DS-5 (Data Protection)	A.14.1.2 (App Security)	CRITICAL	Use prepared queries and enable WAF rules.

TASK 4: DATA CONNECTORS & FASTAPI BACKEND

A high-performance backend API was developed using **FastAPI** (`backend/main.py`). The API accepts log file uploads, coordinates data processing, runs the machine learning model, groups security incidents, and returns structured JSON responses. It includes input validation and error handling to handle invalid files cleanly.

TASK 5: STREAMLIT DASHBOARD & REPORTING

An interactive dashboard was built using **Streamlit** (`app.py`) to provide an intuitive user interface. Security analysts can upload raw log files, view real-time risk metrics, inspect flagged incidents, review compliance controls, and download an executive Markdown security report with one click.

CONCLUSION

The **SecuriAgent AI** project successfully meets all requirements of the Kapstone technical assessment. It demonstrates a practical, working workflow combining machine learning, backend engineering, threat correlation, and compliance reporting in a clean, production-ready codebase.